

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
cat('sdev = 0.025, beta.var.scale = 0.4\n')
```

```
## sdev = 0.025, beta.var.scale = 0.4
```

Key Summary Statistics

```
cat(paste0('```\n', '

--- Summary for K = 1 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.4022  0.4317  0.4394  0.4394  0.4472  0.4833

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.5839  0.7161  0.6637  0.8056  0.9733
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.5667  0.6983  0.6452  0.7839  0.9494
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.01722 0.04500 0.07167 0.23756 0.26556 1.00000
```

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Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.02000 0.04500 0.06944 0.20775 0.20361 1.00000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.01889 0.05222 0.13444 0.33925 0.64000 1.00000

--- Summary for K = 2 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.2244  0.2500  0.2578  0.2584  0.2667  0.2967

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.3311  0.5094  0.4747  0.6394  0.9517
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.3261  0.4967  0.4566  0.6150  0.8783
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01056 0.31167 0.48192 1.00000 1.00000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.004444 0.028333 0.308242 0.817222 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.4811  1.0000  0.7530  1.0000  1.0000

--- Summary for K = 3 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.07444 0.09944 0.10667 0.10766 0.11556 0.17778

```

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beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.1406  0.3944  0.3596  0.5450  0.9339
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.1433  0.3944  0.3514  0.5367  0.8322
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 1.0000  1.0000  0.8364  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0005556 0.0077778 0.2799479 0.6941667 1.0000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.4528  1.0000  0.7480  1.0000  1.0000

--- Summary for K = 4 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.02889 0.04333 0.05056 0.05190 0.05778 0.11444

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.06125 0.32278 0.30242 0.48431 0.87500
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :

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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.06556	0.32500	0.29787	0.47778	0.79167

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6978	0.9372	0.7654	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.002222	0.310012	0.946389	1.000000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	1.0000	1.0000	0.8946	1.0000	1.0000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01278	0.02500	0.02944	0.03099	0.03556	0.09611

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.07444	0.29278	0.27917	0.43500	0.86111

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0800	0.3011	0.2776	0.4367	0.7583

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.6661	1.0000	0.7716	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
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    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.0000  0.2886  0.7772  1.0000
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.9561  1.0000  0.8684  1.0000  1.0000

--- Summary for K = 6 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.008333 0.020556 0.025000 0.026145 0.030556 0.058889

beta.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.02389 0.23389 0.23418 0.37889 0.82444
Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1

gamma.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.02667 0.24333 0.23418 0.38389 0.73000
Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.3822 0.9650 0.6954 1.0000 1.0000
Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.4333 1.0000 1.0000 1.0000 1.0000 1.0000
Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.2340 0.3922 1.0000
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.7139 1.0000 0.7890 1.0000 1.0000

--- Summary for K = 7 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.005556 0.017222 0.022222 0.023237 0.027222 0.070000

beta.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

0.000000 0.007222 0.180556 0.198712 0.334444 0.798889

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.192778	0.199362	0.344444	0.708333

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.6900	0.5593	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2206	1.0000	1.0000	0.9990	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1407	0.0000	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.4106	1.0000	0.7080	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.006111	0.013889	0.017222	0.018442	0.021667	0.052778

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.004444	0.158333	0.186755	0.322222	0.792222

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

```
gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.004444 0.170000 0.188542 0.334444 0.673889
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.0000  0.7150  0.5559  0.9839  1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  1.0000  1.0000  0.9977  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.0000  0.0000  0.1527  0.0000  1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.1994  0.8150  0.6304  1.0000  1.0000
```

--- Summary for K = 9 ---

```
Z.ar:
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.008889 0.023333 0.028333 0.029843 0.034444 0.136111
```

```
beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.008333 0.154444 0.175231 0.291111 0.738333
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
```

```
gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.009028 0.163333 0.173234 0.295556 0.623333
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.0000  0.4178  0.4206  0.8194  1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.9594  1.0000  0.9698  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.0000  0.0000  0.1132  0.0000  1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
```

```
0.0000 0.0000 0.4328 0.4286 0.8117 1.0000
', '\n`'))
```

```
## ' '
##
##
## --- Summary for K = 1 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.4022 0.4317 0.4394 0.4394 0.4472 0.4833
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.5839 0.7161 0.6637 0.8056 0.9733
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.5667 0.6983 0.6452 0.7839 0.9494
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01722 0.04500 0.07167 0.23756 0.26556 1.00000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.02000 0.04500 0.06944 0.20775 0.20361 1.00000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01889 0.05222 0.13444 0.33925 0.64000 1.00000
##
## --- Summary for K = 2 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.2244 0.2500 0.2578 0.2584 0.2667 0.2967
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```



```

## 0.0000 0.3311 0.5094 0.4747 0.6394 0.9517
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.3261 0.4967 0.4566 0.6150 0.8783
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.01056 0.31167 0.48192 1.00000 1.00000
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.000000 0.004444 0.028333 0.308242 0.817222 1.000000
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.4811 1.0000 0.7530 1.0000 1.0000
##
## --- Summary for K = 3 ---
## Z.ar:
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.07444 0.09944 0.10667 0.10766 0.11556 0.17778
##
## beta.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.1406 0.3944 0.3596 0.5450 0.9339
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:

```

```

## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000 0.1433 0.3944 0.3514 0.5367 0.8322
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000 1.0000 1.0000 0.8364 1.0000 1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000000 0.0005556 0.0077778 0.2799479 0.6941667 1.0000000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000 0.4528 1.0000 0.7480 1.0000 1.0000
##
## --- Summary for K = 4 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.02889 0.04333 0.05056 0.05190 0.05778 0.11444
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.00000 0.06125 0.32278 0.30242 0.48431 0.87500
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.00000 0.06556 0.32500 0.29787 0.47778 0.79167
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000 0.6978 0.9372 0.7654 1.0000 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.000000 0.000000 0.002222 0.310012 0.946389 1.000000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
## 0.0000 1.0000 1.0000 0.8946 1.0000 1.0000

```

```

##
## --- Summary for K = 5 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01278 0.02500 0.02944 0.03099 0.03556 0.09611
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.07444 0.29278 0.27917 0.43500 0.86111
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0800 0.3011 0.2776 0.4367 0.7583
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.6661 1.0000 0.7716 1.0000 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.2886 0.7772 1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.9561 1.0000 0.8684 1.0000 1.0000
##
## --- Summary for K = 6 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.008333 0.020556 0.025000 0.026145 0.030556 0.058889
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02389 0.23389 0.23418 0.37889 0.82444
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

```

##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02667 0.24333 0.23418 0.38389 0.73000
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.3822  0.9650  0.6954  1.0000  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.4333  1.0000  1.0000  1.0000  1.0000  1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2340  0.3922  1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.7139  1.0000  0.7890  1.0000  1.0000
##
## --- Summary for K = 7 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.005556 0.017222 0.022222 0.023237 0.027222 0.070000
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.007222 0.180556 0.198712 0.334444 0.798889
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.007778 0.192778 0.199362 0.344444 0.708333
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.6900  0.5593  1.0000  1.0000

```

```

## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.2206  1.0000  1.0000  0.9990  1.0000  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.1407  0.0000  1.0000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.4106  1.0000  0.7080  1.0000  1.0000
##
## --- Summary for K = 8 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.006111 0.013889 0.017222 0.018442 0.021667 0.052778
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.004444 0.158333 0.186755 0.322222 0.792222
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.004444 0.170000 0.188542 0.334444 0.673889
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.7150  0.5559  0.9839  1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  1.0000  1.0000  0.9977  1.0000  1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.1527  0.0000  1.0000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.1994  0.8150  0.6304  1.0000  1.0000
##
## --- Summary for K = 9 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.008889 0.023333 0.028333 0.029843 0.034444 0.136111
##

```

```

## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.008333 0.154444 0.175231 0.291111 0.738333
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.009028 0.163333 0.173234 0.295556 0.623333
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.4178  0.4206  0.8194  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000  0.9594  1.0000  0.9698  1.0000  1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.1132  0.0000  1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.4328  0.4286  0.8117  1.0000
##
## ‘‘‘

```

Conclusion

and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.